

MUNGLINUP WEST, Australia

WMO: 956440

Lat: 33.555S

Lon: 120.700E

Elev: 221

StdP: 98.70

Time Zone: 8.00 (E08)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	4.3	5.0	-2.7	3.1	19.0	-0.8	3.6	18.5	12.4	14.9	11.4	14.3	3.7	320	0.572

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	12.4	35.8	18.8	33.2	18.2	30.7	18.1	21.9	28.6	20.6	27.4	19.6	26.2	5.7	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.0	15.1	24.2	18.7	13.9	22.0	17.9	13.2	20.9	65.1	28.3	60.3	27.7	57.0	26.1	29.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.0	9.8	8.9	2.4	41.3	0.8	1.9	1.9	42.7	1.4	43.8	1.0	44.8	0.4	46.2
			1.7	23.4	0.7	2.0	1.2	24.9	0.7	26.0	0.3	27.2	-0.2	28.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	16.3	20.9	21.0	19.7	17.7	14.5	12.3	11.3	11.9	13.2	15.6	17.8	19.3
	DBStd	4.60	3.47	3.25	3.34	3.20	2.50	2.06	2.05	2.54	3.03	3.48	3.70	3.60
	HDD10.0	25	0	0	0	0	0	3	9	8	4	1	0	0
	HDD18.3	1147	9	9	21	50	121	181	217	199	158	100	54	28
	CDD10.0	2309	338	308	302	231	141	73	51	68	100	175	234	289
	CDD18.3	388	89	83	65	30	3	0	0	1	3	16	38	59
	CDH23.3	3969	904	748	559	239	31	2	1	18	63	243	465	696
	CDH26.7	1751	447	338	237	69	3	0	0	3	17	93	202	342
(14) Wind	WSAvg	4.8	5.3	5.3	4.7	4.4	4.3	4.5	4.6	4.6	4.7	4.7	5.1	5.1
(15) Precipitation	PrecAvg	497	28	34	28	32	52	52	64	62	53	45	39	27
	PrecMax	657	180	115	110	85	130	133	146	159	116	100	109	85
	PrecMin	293	0	1	5	8	10	25	21	15	8	8	3	0
	PrecStd	92	38	30	25	18	30	26	30	31	27	26	23	25
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.0	37.7	36.0	31.7	26.7	22.4	21.4	25.6	29.2	33.4	35.7	38.8
		MCWB	18.9	20.5	18.6	17.0	14.9	13.4	12.6	13.3	15.7	16.4	17.8	17.9
	2%	DB	35.4	34.0	32.5	28.8	23.6	20.1	19.3	21.7	24.8	29.2	31.9	34.1
		MCWB	19.3	19.1	18.1	16.6	14.2	12.8	12.0	12.7	14.5	15.6	16.8	18.2
	5%	DB	31.5	31.0	29.1	26.0	21.5	18.3	17.5	19.1	21.7	25.7	28.6	30.5
		MCWB	19.5	19.4	17.8	16.3	13.7	12.1	11.7	12.2	13.4	15.5	17.0	17.7
	10%	DB	28.1	27.9	26.1	23.7	19.6	16.7	16.0	17.3	19.4	22.7	25.4	26.9
		MCWB	19.0	18.6	17.2	15.8	13.5	11.9	11.1	11.9	12.8	14.9	16.0	17.1
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	25.3	23.6	21.4	19.7	17.5	15.6	14.7	15.4	18.0	20.1	22.1	23.3
		MCDB	30.8	29.5	27.8	25.0	21.2	19.2	18.0	19.8	23.9	26.7	28.6	30.3
	2%	WB	21.9	21.3	20.2	18.6	16.3	14.4	13.5	14.2	15.7	17.9	19.4	20.2
		MCDB	28.5	29.5	26.4	23.2	19.7	17.5	16.6	18.6	21.4	23.9	26.7	27.5
	5%	WB	20.3	20.1	19.3	17.7	15.2	13.5	12.6	13.2	14.4	16.5	18.0	18.7
		MCDB	27.2	27.1	24.7	22.1	18.9	16.4	15.8	17.3	19.3	22.5	24.8	26.4
	10%	WB	19.2	19.2	18.3	16.9	14.3	12.6	11.7	12.3	13.4	15.4	16.8	17.6
		MCDB	25.7	25.7	23.3	21.3	18.0	15.5	14.9	16.2	18.2	21.2	23.3	24.8
(35) Mean Daily Temperature Range	5% DB	MDBR	13.0	12.4	11.8	11.2	9.8	9.0	9.0	10.0	11.6	12.6	13.0	13.5
		MCDBR	19.6	17.9	16.5	14.6	12.8	11.3	11.6	13.7	15.9	18.1	19.0	20.3
		MCWBR	6.3	5.9	5.3	5.6	5.8	5.8	6.1	6.8	6.9	7.4	7.0	6.8
	5% WB	MCDBR	15.4	14.6	13.2	11.4	9.8	9.2	9.7	11.2	13.5	15.0	15.6	16.3
		MCWBR	6.8	5.8	5.3	5.3	5.7	5.8	6.0	6.5	6.9	7.7	7.5	7.3
(40) Clear-Sky Solar Irradiance	taub		0.346	0.342	0.324	0.313	0.298	0.289	0.286	0.295	0.315	0.323	0.330	0.333
	taud		2.543	2.566	2.605	2.628	2.636	2.634	2.644	2.608	2.532	2.536	2.540	2.556
	Ebn at Noon		991	975	958	916	880	863	885	923	952	983	1001	1008
	Edh at Noon		109	103	92	81	71	66	69	80	98	105	109	109
(44) All-Sky Solar Radiation	RadAvg		7.41	6.49	5.37	4.01	3.00	2.55	2.76	3.63	4.85	6.08	6.96	7.67
	RadStd		0.50	0.44	0.35	0.24	0.19	0.10	0.14	0.15	0.18	0.36	0.44	0.49

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (8 neighbors)	+0.23	N/A	N/A	N/A	N/A	N/A	N/A	-60	+93	+44

Nomenclature: See separate page